Computer Networks and Internets with Internet Applications, 4e

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Lecture PowerPoints

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Chapter 23

An Error Reporting Mechanism: Internet Control Message Protocol (ICMP)

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Topics Covered

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23.1 Introduction

This chapter

- examines an error reporting protocol that is integrated with IP
- reviews the basic errors that can be reported
- explains how and where such messages are sent

- ICMP was originally intended
  - to learn why datagrams could not be delivered
- Also can be used to gather information about an internet

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23.2 Best-Effort Semantics And Error Detection

- Error detection in IP?
  - a header checksum that is used to detect transmission errors
  - checksum that covers the entire header
  - checksum is verified to ensure that the header arrived intact

- Checksum error?
  - the datagram must be discarded immediately
  - any fields in the datagram header could be altered

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23.3 Internet Control Message Protocol

- Problems that are less severe than transmission errors result in error conditions that can be reported
- TCP/IP → Internet Control Message Protocol (ICMP)
- Figure 23.1 lists the ICMP messages
  - include both error and informational messages

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Figure 23.1

List of ICMP messages

Each message is identified by an 8-bit type field

Type	Name
0	Echo Reply
1	Unassigned
2	Unassigned
3	Destination Unreachable
4	Source Quench
5	Redirect
6	Alternate Host Address
7	Unassigned
8	Echo
9	Router Advertisement
10	Router Solicitation
11	Time Exceeded
12	Parameter Problem
13	Timestamp
14	Timestamp Reply
15	Information Request
16	Information Reply
17	Address Mask Request
18	Address Mask Reply
19	Reserved (for Security)
20-29	Reserved (for Robustness Experiment)
30	Traceroute
31	Datagram Conversion Error
32	Mobile Host Redirect
33	IPv6 Where-Are-You
34	IPv6 I-Am-Here
35	Mobile Registration Request
36	Mobile Registration Reply
37	Domain Name Request
38	Domain Name Reply
39	SKIP
40	Photuris
41-255	Reserved

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Examples of ICMP Error Messages (1)

- Source Quench
  - A router that run out of buffer space discard datagrams
    - it sends a source quench message to the source (originator)

- Time Exceeded
  - Case 1:
    - If Time To Live (TTL) reach to zero, the router discards the datagram and sends a time exceeded message.
  - Case 2:
    - if the reassembly timer expires before all fragments arrive

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Examples of ICMP Error Messages (2)

- Destination Unreachable
  - If datagram cannot be delivered to its final destination
    - the router sends a destination unreachable message to the source
    - specifies whether the destination host or network is unreachable
- Redirect
  - If a host has incorrectly sent a datagram that should be sent to a different router,
    - the router uses a redirect message to cause the host to change its route (host or network)
- Parameter Problem
  - One of the parameters specified in a datagram is incorrect

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Examples of ICMP Informational Messages

- Echo Request / Reply
  - An echo request message can be sent to any computer
    - Host is required to send an ICMP echo reply message
    - The reply carries the same data as the request
- Address Mask Request / Reply
  - A host broadcasts an address mask request when it boots
    - Routers that receive the request, it sends a 32-bit subnet mask

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23.4 ICMP Message Transport

- The ICMP message is placed in the data area of the IP datagram
- A datagram is encapsulated in a frame for transmission
  - Figure 23.2 illustrates the two levels of encapsulation
- Where should an ICMP message be sent?
- ICMP messages are always created in response to a datagram
  - Either the datagram has encountered a problem
  - Or an ICMP request message to which the destination replies
- An ICMP error message or an ICMP reply is sent back to the source
- ICMP messages do not have special priority
  - They are forwarded like any other datagram
  - But, if an ICMP error message causes an error, no error message is sent, since we don't want network to be congested further

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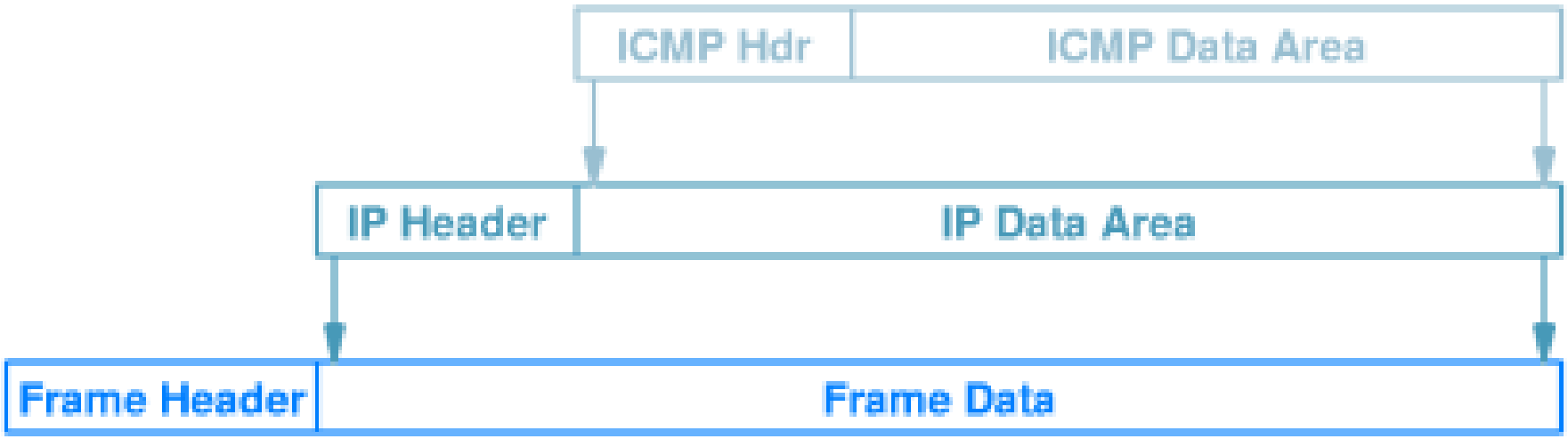

Figure 23.2 Two levels of encapsulation that occur when an ICMP message is sent. The ICMP message is encapsulated in a datagram, which is encapsulated in a frame for transmission across a physical network.

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23.5 Using ICMP Messages To Test Reachability

- Ping uses the ICMP echo request/reply messages
  - Sends an IP datagram that contains an ICMP echo request message to the specified destination
  - After sending the request, it waits a short time for the reply
    - If no reply arrives, ping retransmits the request
    - If no reply arrives for the retransmissions, ping declares that the remote machine is not reachable
- Whenever an echo request arrives, the ICMP software must send an echo reply

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23.6 Using ICMP To Trace A Route

- ICMP messages are used by the "traceroute"
- traceroute constructs a list of all routers along a path to a given destination:
  - sends a series of datagrams and waits for a response to each
  - sets the TTL value in the first datagram to 1
  - The first router that receives the datagram
    - decrements the time to live,
    - discards the datagram,
    - and sends back an ICMP time exceeded message
  - extract the IP source address and announce the address of the first router along the path to the destination

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23.6 Using ICMP To Trace A Route (cont)

- After it discovers the address of the first router, it sends a datagram with TTL set to 2
  - The 1st router decrements the counter and forwards
  - The 2nd router discards and sends an error message
- Similarly, once it has received an error message from a router that is distance 2,
  - it sends a datagram with TTL set to 3 , then 4 , and so on

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23.6 Using ICMP To Trace A Route (cont)

- Several details remain that traceroute must handle
  - Datagrams can be lost, duplicated, or delivered out of order.
- traceroute must be prepared to handle duplicate responses and to retransmit lost datagrams
- Choosing a retransmission timeout can be difficult
  - because traceroute cannot know how long to wait for a response
  - traceroute allows the user to decide how long to wait
- traceroute faces another problem
  - routes can change dynamically
  - If routes change between two probes, the second probe may take a longer/shorter path than the first
- The sequence of routers found may not correspond to a valid path
  - traceroute is most useful in an internet where routes are stable

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23.7 The Last Address Printed By Traceroute

- TTL incremented until datagram to reach its destination
- traceroute sends a datagram, and destination host is required to respond, there are two possibilities
  - Send an ICMP echo request message;
    - the destination host will generate an ICMP echo reply .
  - Send a datagram to a nonexistent application;
    - the destination host will generate an ICMP destination unreachable message

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23.8 Using ICMP For Path MTU Discovery

- Fragmentation solves the problem of heterogeneous networks
  - but it often has negative impact on the performance
- Fragmentation can be avoided if a smaller datagram size selected
- The smallest MTU along a path from a source to a destination is known as the path MTU
  - if routes change, the path MTU can change as well
- What mechanism can a host use to determine the path MTU?
  - It sends a sequence of probes, header bit set to prevent fragmentation
  - If a datagram is larger than the MTU, the router will discard the datagram and send the appropriate ICMP message to the host
  - The host can then send a smaller probe until one succeeds
  - Retransmit probes for which no response is received

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